Household personas



Real-survey data

36 coded fields

Household fields

- Demographics
- Income
- Assets
- Attitudes
- Household
- Spending
- Debt

Construct & calibrate

500M Synthetic personas



Sample without replacement



12M Distinct households

Post-construction PCA
14 ordinal fields

Decision policy learning

4.17M survey QA
Underlying survey QA

2.93M train
Full-data SFT + RLCD

YN-mini
Full-data policy

10 fixed-budget variants

50k / 200k
5 encoder + 5 decoder
SFT / RLCD / CE - Head-only / LoRA + head

8 financial task families

- Income & consumption
- Medical funding
- Purchase funding
- Loan terms

MLX runtime

- Liquid & risky assets
- Multiple debts
- Repayment difficulty
- Liquidity shortfall

Conditions and available options

Genese



State

World / ECS

ID	Cash	Debt
i	c(i)	d(i)
...	...	...
n	c(n)	d(n)

Authoritative state

Requests

State + options

Choice intents

Brain



Option space

a1	p1
a2	p2
a3	p3

YN-mini-MLX

Validate intents → Apply registered effects → Commit state / events

Independent financial ledger



- Consumption
- Closing cash
- Debt payments
- Shortfalls

Execute selected plans

Population-scale experiment

12M households × 2 conditions

24M choices



- Condition A
- Condition B

Choose once / condition



12-month ledger

Purchase goals + approved limits vary together

Fixed-panel monthly experiment

10k × 2 goals × 6 paths × 24 months

2.88M logical choices

Month t state
Cash / debt
Remaining goal

Choose action

Update ledger

Next month t + 1

Repayment + arrears update next credit offers

Household consumption-saving

HANK-inspired household block

1,060 households × 2 paths × 4 months

8,480 decisions / rule



33 feasible options

Rational benchmark

Standalone YN-Lite-4B / YN-master-9B

Advised YN-Lite-4B / YN-master-9B

Consumption + cash

Paired baseline / tax-cut income · Sep-Dec